SUBJECTIVE TYPE

1. Calculate the gravitational field intensity at the centre of the base of a hollow hemispherical shell of mass M and radius R. (Assume the base of hemisphere to be open)

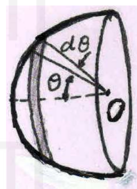
Solution Consider the shaded elemental ring of the hemisphere. Its area is

$$dA = 2\pi(R\sin\theta)(R\,d\theta) = 2\pi R^2 \sin\theta\, d\theta$$

and its mass is

$$dM = \frac{M}{2\pi R^2}dA = \frac{M}{2\pi R^2}\times 2\pi R^2 \sin\theta\, d\theta = M\sin\theta\, d\theta$$

The gravitational field due to this ring at O along the axis is



$$dg = \frac{GdM}{R^2}\cos\theta = \frac{GM}{R^2}\sin\theta\cos\theta\, d\theta$$

The net gravitational field at O is

$$g = \int_0^{\pi/2}\frac{GM}{R^2}\sin\theta\cos\theta\, d\theta = \boxed{\frac{GM}{2R^2}}$$

2. Calculate the gravitational field intensity and potential at the centre of the base of a solid hemisphere of mass M and radius R.

Solution Consider a hemispherical shell of radius r and thickness dr. Its volume is $dV = 2\pi r^2 dr$ and its mass is

$$dM = \frac{M}{(2/3)\pi R^2}dV = \frac{3M}{2\pi R^2}\times 2\pi r^2 dr = \frac{3M}{R^2}r^2 dr$$

The gravitational field due to this shell at the centre of base is

$$dg = \frac{GdM}{2r^2} = \frac{G}{2r^2}\times\frac{3M}{R^2}r^2 dr = \frac{3GM}{2R^2}dr$$

The net gravitational field at the centre is

$$g = \int_0^R \frac{3GM}{2R^2}dr = \boxed{\frac{3GM}{2R}}$$

The gravitational potential being scalar can be calculated by simply considering complete solid sphere of mass $2M$ and radius R. The gravitational potential due to this complete sphere at the centre is

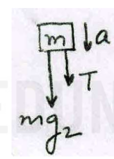
$$-\frac{3}{2}\times\frac{G(2M)}{R} = -\frac{3GM}{R}$$

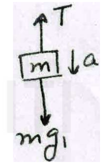
The gravitational potential due to solid hemisphere of mass M and radius R is half of the value obtained above and is given by

$$\boxed{V = -\frac{3GM}{2R}}$$

3. Two point objects, each of mass m, are connected by a massless rope of length l. The objects are held near the surface of Earth, so that the rope is vertical and one object is hanging below the other. Then the objects are released. If $l <<$ R, show that the tension in the rope is $T = \frac{GM\,ml}{R^3}$ where M is the mass of the Earth and R is its radius.

Solution Let h be the height of lower mass from the surface of earth. Then the height of upper mass is $h + l$. Let g_0 be the acceleration due to gravity on the surface of earth, and let g_1 and g_2 be acceleration due to gravity at the site of lower and upper masses respectively.





Then $\quad g_1 = g_0\left(1-\frac{2h}{R}\right)$

and $\quad g_2 = g_0\left(1-\frac{2(h+l)}{R}\right)$

$$\Rightarrow\ g_1 - g_2 = 2g_0 l/R.$$

From FBD, $\quad ma = mg_1 - T \quad$ and $\quad ma = T + mg_2$

$$\Rightarrow\ mg_1 - T = T + mg_2$$

$$\therefore\ T = \frac{m}{2}(g_1-g_2) = \frac{m}{2}\times\frac{2l}{R}g_0 = \frac{ml}{R}\times\frac{GM}{R^2} = \boxed{\frac{GMml}{R^3}}$$

4. A small satellite revolves around a heavy planet in a circular orbit. At certain point in its orbit, a sharp impulse acts on it and instantaneously increases its kinetic energy to k (<2) times without change in its direction of motion. Show that in its subsequent motion, the ratio of its maximum and minimum distances from the planet is $\frac{k}{2-k}$, assuming the mass of the satellite to be negligibly small as compared to that of the planet.

Solution Let the satellite of mass m revolves around the planet of mass M in an orbit of radius r. Then, orbital velocity is

$$v_0 = \sqrt{\frac{GM}{r}}$$

As its kinetic energy is increased k (< 2) times, it goes into an elliptical orbit. Let the maximum (or minimum) distance of the satellite from the planet be x and its speed be v. Then, from conservation of angular momentum, we have

$$m\sqrt{\frac{kGM}{r}}\times r = m v x \qquad .....(1)$$

Also, from conservation of mechanical energy, we have

$$\frac{-GMm}{r} + \frac{1}{2}m\left(\sqrt{\frac{kGM}{r}}\right)^2 = \frac{-GMm}{x} + \frac{1}{2}mv^2 \qquad ....(2)$$

Using value of v from equation (1), we get

$$\frac{-GMm}{r} + \frac{1}{2}m\frac{kGM}{r} = \frac{-GMm}{x} + \frac{1}{2}m\frac{r^2}{x^2}\left(\frac{kGM}{r}\right)$$

$$\Rightarrow\ \frac{-1}{r} + \frac{k}{2r} = \frac{-1}{x} + \frac{kr}{2x^2}$$

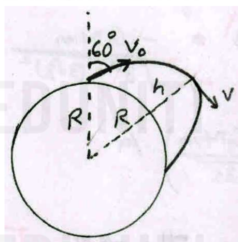
$$\Rightarrow\ (2-k)\left(\frac{x}{r}\right)^2 - 2\left(\frac{x}{r}\right) + k = 0$$

$$\Rightarrow \frac{x}{r}=\frac{2\pm\sqrt{2^2-4k(2-k)}}{2(2-k)}=\frac{1\pm(k-1)}{2-k}=\frac{k}{2-k},1$$

$$\Rightarrow x_{max}=\frac{kr}{2-k} \quad \text{and} \quad x_{min}=r$$

$$\therefore \boxed{\frac{x_{max}}{x_{min}}=\frac{k}{2-k}}$$

5. A body is launched from the Earth's surface at an angle $\alpha=30^0$ to the horizontal at a speed $v_0=\sqrt{1.5GM/R}$
Neglecting air resistance and Earth's rotation, find
(a) the height to which the body will rise.
(b) the radius of curvature of trajectory at its top point.
Solution
(a) Let h be the height from the earth's surface to which the body will rise. The angle of projection to the radius vector of Earth is 60^o. Form conservation of angular momentum, we have



$$mv_0 R\sin 60^o = mv(h+R)$$

$$\Rightarrow v=\frac{\sqrt{3}\,v_0 R}{2(h+R)}=\sqrt{\frac{9}{8}\frac{GMR}{(h+R)^2}}$$

Also, from conservation of mechanical energy, we have

$$\Rightarrow \frac{-GMm}{R}+\frac{3}{4}\frac{GMm}{R}=\frac{-GMm}{R+h}+\frac{1}{2}m\left(\frac{9}{8}\frac{GMR}{(h+R)^2}\right)$$

$$\Rightarrow \frac{-1}{4R}=\frac{-1}{R+h}+\frac{9R}{16(h+R)^2}$$

$$\Rightarrow 4(h+R)^2-16(h+R)R+9R^2=0$$

$$\Rightarrow \frac{h+R}{R}=\frac{16\pm\sqrt{6^2-4\times4\times9}}{8}$$

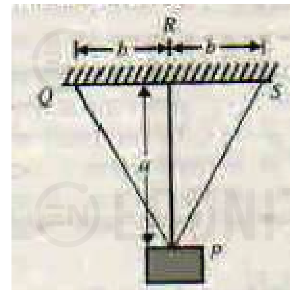
$$\Rightarrow \frac{h}{R}+1=2\pm\frac{\sqrt{7}}{2}$$

Ignoring negative value, we get $\boxed{h=(1+\sqrt{7}/2)R}$

(b) The acceleration of body at the top point is $a=\dfrac{GM}{(R+h)^2}$
If r is the radius of curvature of trajectory at its top point, then $a = v^2/r$

$$\therefore r=\frac{v^2}{a}=\frac{9}{8}\frac{GMR}{(R+h)^2}\Big/\frac{GM}{(R+h)^2}=\boxed{\frac{9}{8}R}$$

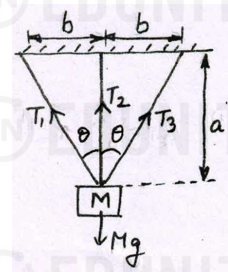
6. Three elastic wires, PQ, PR and PS support a body P of mass M, as shown in fig. The wires are of the same material and cross-sectional area. The middle wire is vertical. Find the loads carried by each wire.



Solution By symmetry, $T_1=T_3$

In equilibrium, $2T_1\cos\theta+T_2=Mg$ .....(1)

Also, vertical deflection in the middle-wire and side-wire must be equal



$$\Rightarrow \Delta\ell_1\cos\theta=\Delta\ell_2$$

$$\Rightarrow \frac{T_1\ell_1}{AY}\cos\theta=\frac{T_2\ell_2}{AY}$$

$$\Rightarrow T_1\ell_1\cos\theta=T_2\ell_2$$

where $\ell_2=\ell_1\cos\theta$

$$\Rightarrow T_1=T_2$$

From eqn (1), $2T_1\cos\theta+T_1=Mg$

$$\Rightarrow T_1=\frac{Mg}{2\cos\theta+1}=\frac{Mg}{2(b/\sqrt{a^2+b^2})+1}$$

$$\therefore \boxed{T_1=T_2=T_3=\frac{\sqrt{a^2+b^2}\,Mg}{2b+\sqrt{a^2+b^2}}}$$

PARAGRAPH TYPE

Paragraph for Question Nos. 7 to 9

A non-homogeneous sphere of radius R has the following density variation

$\rho=\rho_0$ for $r\le R/3$

$\rho=\dfrac{\rho_0}{2}$ for $\dfrac{R}{3}<r\le\dfrac{3R}{4}$ and

$\rho=\dfrac{\rho_0}{8}$ for $\dfrac{3R}{4}<r\le R$

7. The gravitational field due to sphere at $r=R/4$ is

(a) $\dfrac{\pi GR\rho_0}{3}$ (b) $\dfrac{2\pi GR\rho_0}{3}$

(c) $\dfrac{2\pi GR\rho_0}{9}$ (d) $\dfrac{\pi GR\rho_0}{16}$

8. The gravitational field due to sphere at $r = R/2$ is

(a) $\dfrac{43\pi G R\rho_0}{50}$ (b) $\dfrac{23\pi G R\rho_0}{100}$

(c) $\dfrac{18\pi G R\rho_0}{25}$ (d) $\dfrac{35\pi G R\rho_0}{81}$

9. The gravitational field due to sphere at $r = 5R/6$ is

(a) $0.98\pi G R\rho_0$ (b) $0.48\pi G R\rho_0$

(c) $0.23\pi G R\rho_0$ (d) $0.52\pi G R\rho_0$

Solution

7 (a) The gravitational field at r = R/4 is due to the mass M contained in the sphere of radius R/4.

Here, $M = \rho_0 \times \frac{4}{3}\pi\left(\frac{R}{4}\right)^3 = \frac{\rho_0 \pi R^3}{48}$

$$\therefore \quad g = \frac{GM}{(R/4)^2} = \frac{G\rho_0 \pi R^3/48}{R^2/16} = \frac{\pi G R \rho_0}{3}$$

8 (d) The mass contained in sphere of radius r = R/2 is

$$M = \rho_0 \times \frac{4}{3}\pi\left(\frac{R}{3}\right)^3 + \frac{\rho_0}{2}\times\frac{4}{3}\pi\left[\left(\frac{R}{2}\right)^3 - \left(\frac{R}{3}\right)^3\right]$$

$$= \frac{4}{3}\pi\rho_0 R^3\left(\frac{1}{27} + \frac{1}{16} - \frac{1}{54}\right) = \frac{35\pi\rho_0 R^3}{324}$$

$$\therefore \quad g = \frac{GM}{(R/2)^2} = \frac{G\times 35\pi\rho_0 R^3/324}{R^2/4} = \frac{35\pi G R\rho_0}{81}$$

9 (b) The mass contained in sphere of radius r = 5R/6 is

$$M = \rho_0 \times \frac{4}{3}\pi\left(\frac{R}{3}\right)^3 + \frac{\rho_0}{2}\times\frac{4}{3}\pi\left[\left(\frac{3R}{4}\right)^3 - \left(\frac{R}{3}\right)^3\right]$$

$$+\frac{\rho_0}{8}\times\frac{4}{3}\pi\left[\left(\frac{5R}{6}\right)^3 - \left(\frac{3R}{4}\right)^3\right] = \frac{3443\pi\rho_0 R^3}{81\times 128}$$

$$\therefore \quad g = \frac{GM}{(5R/6)^2} = \frac{3443\pi\rho_0 R^3/(81\times128)}{25R^2/36} = 0.48\pi G R\rho_0$$

Paragraph for Question Nos. 10 to 12

It can be assumed that orbits of earth and mars are nearly circular around the sun. It is proposed to launch an artificial planet around the sun such that its apogee is at the orbit of mars, while its perigee is at the orbit of earth. Let T_e and T_m be periods of revolution of earth and mars. Further the variables are assigned the meanings as follows.

M_e, M_m = Masses of earth and mars

L_e, L_m = Angular momenta of earth and mars (each about the sun)

a_e, a_m = Semi-major axes of earth and mars orbits

m = Mass of the artificial planet

E_e, E_m = Total energies of earth and mars

10. Time period of revolution of the artificial planet about the Sun will be (neglect gravitational effects of Earth and Mars)

(a) $\dfrac{T_e + T_m}{2}$ (b) $\sqrt{T_e T_m}$

(c) $\dfrac{2T_e T_m}{T_e + T_m}$ (d) $\left(\dfrac{T_m^{2/3} + T_e^{2/3}}{2}\right)^{3/2}$

11. The total energy of artificial planet's orbit will be

(a) $\dfrac{2m}{M_e}\left(\dfrac{a_e E_e}{a_e + a_m}\right)$ (b) $\dfrac{2m}{M_m}\left(\dfrac{a_e E_e}{a_e + a_m}\right)$

(c) $\dfrac{2E_e m}{M_m}\left(\dfrac{a_e + a_m}{a_m}\right)$ (d) $\dfrac{2E_e m}{M_e}\left(\dfrac{a_e + a_m}{\sqrt{a_e^2 + a_m^2}}\right)$

12. Areal velocity of the artificial planet around the sun will be

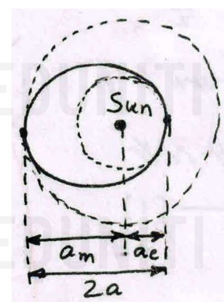
(a) less than that of earth (b) more than that of mars

(c) more than that of earth (d) same as that of earth

Solution

10(d) Let T and a be the time period and semi major axis of orbit of the artificial planet Then,

$T^2 \,\alpha\, a^3 \quad \Rightarrow \quad a = c\,T^{2/3}$



Similarly, $a_m = c\,T_m^{2/3}$

and $a_e = c\,T_e^{2/3}$

Also, $2a = a_m + a_e$

$\Rightarrow \quad 2c\,T^{2/3} = c\,T_m^{2/3} + c\,T_e^{2/3}$

$$\therefore \quad T = \left(\frac{T_m^{2/3} + T_e^{2/3}}{2}\right)^{3/2}$$

11(a) If M_s is mass of sun and E is total energy of artificial planet, then

$$E = -\frac{GM_s m}{2a} \text{ and } E_e = -\frac{GM_s M_e}{2a_e}$$

$$\Rightarrow \quad \frac{E}{E_e} = \frac{m}{M_e}\times\frac{2a_e}{2a} = \frac{2m}{M_e}\left(\frac{a_e}{a_e + a_m}\right)$$

$$\therefore \quad E = \frac{2m}{M_e}\left(\frac{a_e E_e}{a_e + a_m}\right)$$

12 (c) Areal velocity of planet around the sun is

$$\frac{dA}{dt} = \frac{\pi ab}{T} = \frac{\pi a\times a\sqrt{1-e^2}}{\sqrt{4\pi^2\times a^3/GM}} = \sqrt{\frac{GM_s a}{4}}\sqrt{1-\left(\frac{a_m - a_e}{a_m + a_e}\right)^2}$$

$$= \sqrt{\frac{GM_s a}{4}}\sqrt{\frac{4a_m a_e}{2a\,(a_m + a_e)}} = \sqrt{\frac{GM_s}{2(1/a_m + 1/a_e)}}$$

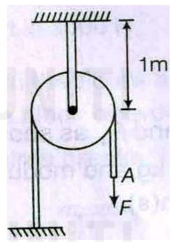
$$\left(\frac{dA}{dt}\right)_e = \sqrt{\frac{GM_s}{2(1/a_e + 1/a_e)}} \quad \text{and}$$

$$\left(\frac{dA}{dt}\right)_m = \sqrt{\frac{GM_s}{2(1/a_m + 1/a_m)}}$$

Now, $\dfrac{1}{a_e} > \dfrac{1}{a} > \dfrac{1}{a_m}$ $\quad\therefore\quad \left(\dfrac{dA}{dt}\right)_e < \dfrac{dA}{dt} < \left(\dfrac{dA}{dt}\right)_m$

Paragraph for Question Nos. 13 to 15

The axle of a pulley of mass 1 kg is attached to the end of an elastic string of length 1 m, cross-sectional area 10^{-3} m^3 and Young's modulus 2×10^5 Nm^{-2}, whose other end is fixed to the ceiling. An inextensible rope of negligible mass is placed on the pulley such that its left end is fixed to the ground and its right end is hanging freely, from the pulley which is at rest in equilibrium. Now, the free end A of the rope is pulled with constant force F = 10 N. Friction between the rope and the pulley can be neglected. (Given, g = 10 ms^{-2})



13. The elongation of the string before applying force is

(a) 50 cm (b) 5 cm
(c) 0.5 cm (d) 0.05 cm

14. The maximum elongation of the string is

(a) 35 cm (b) 30 cm
(c) 25 cm (d) 20 cm

15. The maximum displacement of point A after applying F is

(a) 70 cm (b) 60 cm
(c) 40 cm (d) 50 cm

13(b) Before applying force,

$$\Delta L = \frac{mg\,L}{AY} = \frac{1\times10\times1}{10^{-3}\times2\times10^5} = 5\times10^{-2}\,m = 5\ cm$$

14(c) Force constant,

$$k = \frac{mg}{\Delta L} = \frac{AY}{L} = \frac{10^{-3}\times2\times10^5}{1} = 200\ N/m$$

When the elongation is maximum, pulley is at instantaneous rest. Let the additional elongation at this instant be x. Then,

$$W_{mg} + W_F = \Delta K + \Delta U$$

$$\Rightarrow\ mgx + F\times2x = 0 + \frac{1}{2}k\,[(x+\Delta L)^2 - (\Delta L)^2]$$

$$\Rightarrow\ 1\times10x + 10\times2x = \frac{200}{2}[x^2 + 2x\ \Delta L]$$

$$\Rightarrow\ 30x = 100\,(x^2 + 2x\times0.05)$$

$$\Rightarrow\ 100x^2 - 20x = 0 \qquad \Rightarrow\ x = 0.2\,m = 20\ cm$$

Maximum elongation, $x_m = x + \Delta L = 20 + 5 = 25$ cm.

15(c) The maximum displacement of point A after applying F is

$$2x = 2\times20 = 40\ cm$$

MORE THAN ONE CORRECT

16. Two satellites S_1 and S_2 are respectively into the orbits of heights $R/4$ and $R/6$ above the surface of earth of radius R. They revolve around the earth in coplanar circular orbits in opposite sense. Which of the following statement(s) is/are correct?

(a) The ratio of their time periods is $(15/14)^{3/2}$
(b) The ratio of their time periods is $(14/15)^{3/2}$
(c) The ratio of their speeds of projection form the surface of earth is $\sqrt{23/15}$
(d) The ratio of their speeds of projection form the surface of earth is $\sqrt{21/20}$

16 (a, d) The orbital radius of S_1 and S_2 are respectively $r_1 = R + R/4 = 5R/4$ and $r_2 = R + R/6 = 7R/6$

$$\frac{T_1}{T_2} = \left(\frac{r_1}{r_2}\right)^{3/2} = \left(\frac{5R/4}{7R/6}\right)^{3/2} = \left(\frac{15}{14}\right)^{3/2}$$

If v is the projection speed, then by conservation of mechanical energy,

$$\frac{1}{2}mv^2 - \frac{GMm}{R} = -\frac{GMm}{2r} \qquad v = \sqrt{GM\left(\frac{2}{R} - \frac{1}{r}\right)}$$

$$v_1 = \sqrt{GM\left(\frac{2}{R} - \frac{1}{5R/4}\right)} = \sqrt{\frac{6GM}{5R}} \quad \text{and}$$

$$v_2 = \sqrt{GM\left(\frac{2}{R} - \frac{1}{7R/6}\right)} = \sqrt{\frac{8GM}{7R}}$$

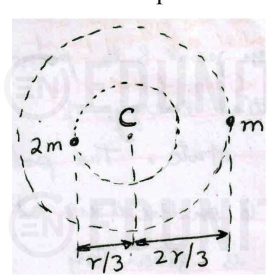
$$\therefore\ \frac{v_1}{v_2} = \frac{\sqrt{6/5}}{\sqrt{8/7}} = \sqrt{\frac{21}{20}}$$

17. A double star is a system of two stars of masses m and 2m, rotating about their centre of mass only under their mutual gravitational attraction. If r is the separation between these two stars, then their time period of rotation about their centre of mass will be proportional to

(a) $r^{3/2}$ (b) r
(c) $m^{1/2}$ (d) $m^{-1/2}$

17 (a, b) Both stars revolve around their common centre of mass C, in circular orbit with same time period. Consider any star, say star of mass m.

The inward gravitational force on it must be equal to the outward centrifugal force.



$$\Rightarrow\ \frac{G(2m)m}{r^2} = m\left(\frac{2\pi}{T}\right)^2\left(\frac{2r}{3}\right)$$

$$\Rightarrow\ T = \sqrt{\frac{4\pi^2 r^3}{3Gm}}$$

$$\therefore\ T \propto r^{3/2} \text{ and } T \propto m^{-1/2}$$

18. A double star consists of two stars having masses M and 2M. The distance between their centres is equal to r. They re-

volve under their mutual gravitational interaction. Then, which of the following statements are correct

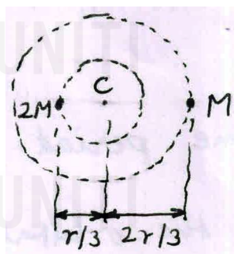
(a) Heavier star revolves in orbit of radius r/3

(b) Both the stars revolve with the same speed, period of which is equal to $(2\pi/r^3)(2GM^2/3)$

(c) Kinetic energy of the lighter star is twice that of the other star.

(d) none of these

18 (a, c) Both stars revolve around their common centre of mass C in circular orbit with same time period. The orbital radius of heavier star is r/3. The lighter star has to cover larger distance in same time. Hence, its speed is more. As its orbital radius is double, it has to cover double the distance and hence, its speed is double. As $K = (1/2)\,mv^2$, the mass of lighter star is half and speed is double, its KE is twice that of the heavier star.



19. A planet of mass m is moving in an elliptical orbit with the sun of mass M at its focus. If the perihelion and aphelion distances are r_2 and r_1 respectively, choose the correct option(s).

(a) Angular momentum of planet relative to sun is $L = \sqrt{\dfrac{2GMm^2\ r_1 r_2}{r_1 + r_2}}$

(b) Total mechanical energy depends only on the semi-major axis of the orbit

(c) Total mechanical energy depends on both semi-major and semi-minor axes of the orbit

(d) $\dfrac{L^2}{2m}\left(\dfrac{1}{r_1^2} - \dfrac{1}{r_2^2}\right) = GMm\left(\dfrac{1}{r_1} - \dfrac{1}{r_2}\right)$

19 (a, b, d) $L = mv_1 r_1 = mv_2 r_2$ and

$$E = \frac{-GMm}{r_1} + \frac{1}{2}mv_1^2 = \frac{-GMm}{r_2} + \frac{1}{2}mv_2^2$$

$$\Rightarrow \frac{-GMm}{r_1} + \frac{1}{2}m\left(\frac{L}{m\,r_1}\right)^2 = \frac{-GMm}{r_2} + \frac{1}{2}m\left(\frac{L}{m\,r_2}\right)^2$$

$$\therefore \frac{L^2}{2m}\left(\frac{1}{r_1^2} - \frac{1}{r_2^2}\right) = GMm\left(\frac{1}{r_1} - \frac{1}{r_2}\right)$$

$$\therefore L = \sqrt{\frac{2GMm^2\, r_1 r_2}{r_1 + r_2}}$$

Total mechanical energy is $E = -GMm/2a$ and depends only on semi - major axis.

20. An earth's satellite is revolving in a circular orbit of radius a with a velocity v_0. A gun is in the satellite and is aimed directly towards the earth. A bullet is fired from the gun with a muzzle velocity $v_0/2$. Neglecting resistance offered by cosmic dust and recoil of gun during subsequent motion of bullet from the centre of the earth, choose the correct option(s).

(a) maximum distance of bullet is 3a

(b) minimum distance of bullet is $3a/4$

(c) maximum distance of bullet is 2a

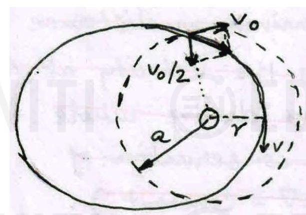
(d) minimum distance of bullet is $2a/3$

20 (c,d) The speed of bullet w.r.t. earth is $\sqrt{v_0^2 + (v_0/2)^2} = \sqrt{5}\,v_0/2$

By cons. of angular momentum and energy,

$$m v_0 a = m v r \text{ and } \frac{1}{2}m\left(\frac{\sqrt{5}}{2}v_0\right)^2 - \frac{GMm}{a} = \frac{1}{2}mv^2 - \frac{GMm}{r}$$

$$\Rightarrow GM\left(\frac{1}{r} - \frac{1}{a}\right) = \frac{1}{2}\times\left(v_0\frac{a}{r}\right)^2 - \frac{1}{2}\times\frac{5}{4}v_0^2$$

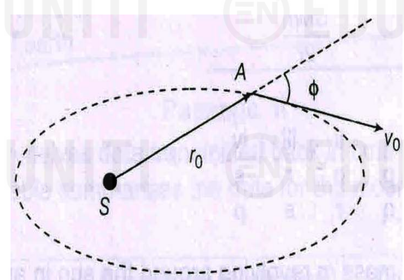


$$\Rightarrow \frac{GM(a-r)}{ar} = \frac{1}{2}\left(\frac{a^2}{r^2} - \frac{5}{4}\right)\times\frac{GM}{a}$$

$$\Rightarrow 3r^2 - 8ar + 4a^2 = 0 \qquad \Rightarrow (r-2a)\,(3r-2a) = 0$$

Therefore, r = 2a/3 is minimum distance and r = 2a is maximum distance

21. A planet A at a certain moment of time was at a distance r_0 from the sun and was travelling with a velocity v_0. The angle between r_0 and v_0 is ϕ. If the mass of sun is M and , $K = \dfrac{r_0^2 v_0^2}{GM}$, then the



(a) minimum distance of the planet from the sun is $\dfrac{r_0}{2-K}(1 - \sqrt{1 + K(2-K)\sin^2\phi})$

(b) minimum distance of the planet from the sun is $\dfrac{r_0}{2-K}(1 - \sqrt{1 - K(2-K)\sin^2\phi})$

(c) maximum distance of the planet from the sun is

$\frac{r_0}{2-K}(1+\sqrt{1-K(2-K)\sin^2\phi})$

(d) maximum distance of the planet from the sun is $\frac{r_0}{2-K}(1+\sqrt{1+K(2-K)\sin^2\phi})$

21(b, c) From conservation of angular momentum,

$mv_0\, r_0 \sin\phi = mvr$ .....(1)

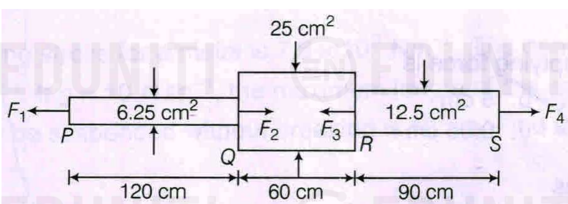
From conservation of mechanical energy,

$\frac{1}{2}mv_0^2 - \frac{GMm}{r_0} = \frac{1}{2}mv^2 - \frac{GMm}{r}$ .....(2)

Solving (1) and (2), we get

$r = \frac{r_0}{2-K}(1 \pm \sqrt{1-K(2-K)\sin^2\phi})$ where $K = \frac{r_0^2 v_0^2}{GM}$

22. A member PQRS is subjected to point loads F_1, F_2, F_3 and F_4 as shown in the figure. Given, $F_1 = 4500$ kg, $F_3 = 45000$ kg, $F_4 = 13000$ kg and modulus of elasticity $= 2.1\times10^6$ kg cm^2. Choose the correct option(s)



(a) For equilibrium, $F_2 = 36500$ kg
(b) Total elongation is 0.98 mm
(c) For equilibrium, $F_2 = 26500$ kg
(d) Total elongation is 0.49 mm

22 (a, d) For equilibrium, Tension in left part is $F_1 = 4500$ N and elongation is

$\frac{FL}{AY} = \frac{4500\times1.2}{6.25\times10^{-4}\,Y} = \frac{8.64\times10^6}{Y}$

Tension in right part is $F_4 = 13000$ N and elongation is

$\frac{FL}{AY} = \frac{13000\times0.9}{12.5\times10^{-4}\,Y} = \frac{9.36\times10^6}{Y}$

Compression in middle part is $F_3 - F_4 = 45000 - 13000 = 32000$ N and compression is

$\frac{FL}{AY} = \frac{32000\times0.6}{25\times10^{-4}\,Y} = \frac{7.68\times10^6}{Y}$

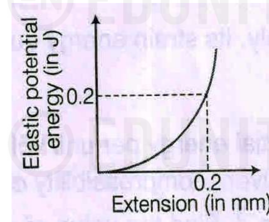
Total elongation is

$\frac{(8.64+9.36-7.68)\times10^6}{Y} = \frac{10.32\times10^6}{2.1\times10^6/10^{-4}} = 4.9\times10^{-4}$ m

$= 0.49$ mm

23. Figure shows the graph of elastic potential energy (U) stored versus extension, for a steel wire ($Y = 2\times10^{11}$ Pa) of volume 200 cc. If area of cross-section is A and original length is L, then

(a) $A = 10^{-4}\ m^2$
(b) $A = 10^{-3}\ m^2$
(c) $L = 1.5$ m
(d) $L = 2$m



23(a, d) $U = \frac{1}{2}\left(\frac{AY}{L}\right)(\Delta L)^2$

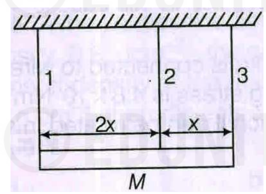
$\Rightarrow \frac{A}{L} = \frac{2U}{Y(\Delta L)^2} = \frac{2\times0.2}{2\times10^{11}\times(0.2\times10^{-3})^2} = 5\times10^{-5}$ m ..(1)

Volume $= AL = 200cc = 2\times10^{-4}\ m^3$ ..(2)

From (1) and (2), $A = 10^{-4}\ m^2$ and $L = 2$m.

24. Three vertical wires 1, 2 and 3 are supporting a rod of mass M in horizontal position as shown in the figure. The wires are of equal length and equal cross-sectional area. It is given that $Y_2 = Y_3$ and the rod remains horizontal always. The wires 1 and 3 are attached at the exterme ends of the rod. Choose the correct option(s)

(a) $T_1 = 2T_3$
(b) $T_1 = 4T_3/3$
(c) $T_2 = T_3$
(d) $T_1 = \frac{2Mg}{5}$

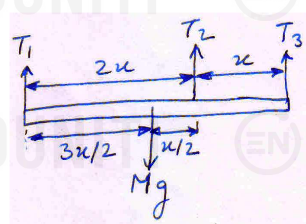


24(b, c, d) $Y = \frac{TL}{A\Delta L}$

L and A are same for all wires. As the rod remains horizontal always, ΔL is also same for all wires.

$\Rightarrow$ $T \propto Y$.

As $Y_2 = Y_3$, we have $T_2 = T_3$



From rotational equilibrium about the centre,

$T_1(3x/2) = T_2(x/2) + T_3(3x/2)$

$\Rightarrow$ $3T_1 = T_2 + 3T_3 = 4T_3$ $\therefore$ $T_1 = 4T_3/3$

From translational equilibrium,

$Mg = T_1 + T_2 + T_3 = T_1 + 2T_3 = T_1 + 2\times3T_1/4 = 5T_1/2$

$\therefore$ $T_1 = 2Mg/5$